

MANIKANTA NEERUKATTU (MANI N)

Senior Software Engineer | Applied AI Engineer | .NET, Azure, AWS & Agentic AI Systems

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PROFESSIONAL SUMMARY

Senior Software Engineer with around 10 years of experience in the Microsoft and AWS ecosystems, progressing from full-stack enterprise development into Applied AI Engineering. Specializes in designing and shipping production-grade AI systems — Advanced RAG pipelines, multi-agent orchestration, and LLM-powered automation — that solve real business problems rather than stay as prototypes.

Strong backend foundation in .NET 8, ASP.NET Core, and Azure, complemented by hands-on cloud engineering across AWS services (EC2, S3, Lambda, RDS, DynamoDB, IAM, ECS/EKS) and deep depth in Azure AI Foundry, AWS Bedrock, Semantic Kernel, and agentic frameworks. Has delivered AI-integrated systems across financial services, healthcare, and ERP, with consistent focus on reliability, scalability, and observable production behavior. Currently completing a Master of Science in AI/ML at Liverpool John Moores University, UK.

SKILLS

AI Engineering: Azure AI Foundry · Azure OpenAI (GPT-4, GPT-4o, Embeddings, Function Calling) · AWS Bedrock (Claude, Titan, Llama, Knowledge Bases, Agents for Bedrock) · Semantic Kernel · Microsoft Agent Framework · LangChain · LangGraph · LlamaIndex · AutoGen · CrewAI · Ollama · HuggingFace · Mistral · RAG & Advanced RAG · Hybrid Search · Semantic Re-ranking · Prompt Engineering · LLM Evaluation (RAGAS) · MCP (Model Context Protocol) · Vector Databases (Azure AI Search, pgvector, ChromaDB) · Multi-Agent Orchestration · Agent Memory Architectures

Backend & System Design: C# · .NET 8/10 · ASP.NET Core · Python · FastAPI · Microservices Architecture · Event-Driven Systems · REST APIs · Azure SDK Integration · AWS SDK Integration · Domain-Driven Design · Clean Architecture

Cloud & Infrastructure: Microsoft Azure (Functions · API Management · Blob Storage · Service Bus · Application Insights · Key Vault) · AWS (EC2 · S3 · Lambda · RDS · DynamoDB · IAM · ECS/EKS · Bedrock · CloudWatch · API Gateway · SNS/SQS) · Azure DevOps · GitHub Actions · Docker · OpenTelemetry · Distributed Tracing · Structured Logging

Data & Storage: PostgreSQL · SQL Server · Azure SQL · pgvector · DynamoDB · RDS (MySQL/PostgreSQL) · Azure Data Factory · Databricks · PySpark

Frontend: Angular · React.js · Blazor · TypeScript · RxJS

Testing: xUnit · NUnit · Cypress · Playwright · SonarQube

PROFESSIONAL EXPERIENCE

Senior Software Engineer — AI & Applied Intelligence | Insightsoftware India Pvt Ltd | Hyderabad, India | Jun 2023 – Present

Led the AI transformation of enterprise financial disclosure and reporting systems. Responsible for taking AI features from design through to production, working across the full stack — from Azure and AWS infrastructure to API surfaces to Angular and React frontend integration.

- Designed and built an Advanced RAG pipeline using Azure AI Foundry, Azure OpenAI, and Azure AI Search — combining hybrid retrieval (keyword + vector), semantic re-ranking, intelligent chunking, and query expansion for high-accuracy compliance document processing.
- Integrated AWS Bedrock (Claude and Titan models) as an alternative inference layer, enabling multi-cloud LLM routing and reducing single-vendor lock-in; deployed inference endpoints via AWS Lambda and API Gateway with IAM-governed access control.
- Engineered a multi-agent reporting system using Semantic Kernel, AutoGen, and LangGraph, with specialized agents for data retrieval (SQL/API), LLM reasoning, and structured report generation; reduced manual reporting effort by 60%.

- Built and integrated MCP (Model Context Protocol) enterprise connectors enabling AI agents to interact with SQL databases, REST APIs, and external SaaS systems — supporting natural language queries, dynamic function calling, and real-time schema exploration.
- Implemented memory-driven agentic workflows with vector memory and persistent context, enabling multi-turn document-aware AI interactions across long-horizon financial tasks.
- Developed a multi-model routing layer directing Azure OpenAI, AWS Bedrock, and local models to specialized tasks (reasoning, extraction, summarization) to balance cost, latency, and output quality.
- Built scalable .NET 8 microservices integrating Azure OpenAI and AI Foundry APIs, with clean architecture, resilience patterns, and Azure API Management.
- Developed Angular-based frontend modules for AI-assisted report generation and disclosure workflows, consuming AI service APIs and providing real-time feedback on pipeline progress and output quality.
- Provisioned and managed AWS infrastructure supporting data pipelines: S3 for document storage, RDS (PostgreSQL) for structured data, DynamoDB for agent state and session persistence, and ECS for containerized microservice deployments — all secured via IAM roles and policies.
- Created prompt evaluation frameworks to systematically measure and improve LLM output quality, reducing hallucinations across model upgrades.
- Set up and maintained CI/CD pipelines (Azure DevOps) with automated quality gates including test coverage enforcement and static analysis.
- Instrumented services with OpenTelemetry, distributed tracing, and structured logging, providing operational visibility across distributed AI workflows spanning Azure and AWS.

Tech Stack: Azure AI Foundry · Azure OpenAI · AWS Bedrock · Semantic Kernel · LangGraph · AutoGen · MCP · .NET 8 · Azure Functions · AWS Lambda · AWS ECS · AWS S3 · AWS RDS · AWS DynamoDB · AWS IAM · Azure AI Search · pgvector · Angular · React · Azure DevOps

Software Technologist | Philips | Bangalore, India | Apr 2022 – Jun 2023

Delivered enhancements to a regulated healthcare platform, integrating AI capabilities and strengthening API security and reliability.

- Integrated Azure Cognitive Services and Azure OpenAI to enable intelligent document processing and clinical decision support features within the platform.
- Built secure, high-throughput APIs using ASP.NET Core and Azure API Management, with OAuth2/OIDC authentication and input validation aligned to healthcare data compliance requirements.
- Developed real-time communication modules (chat and video) using SignalR, integrated into patient and clinician-facing web applications.
- Established automated regression testing pipelines using Cypress, cutting manual QA cycles and improving release stability.

Tech Stack: .NET Core · React · Angular · Azure Services · Azure OpenAI · SQL Server · Cypress

Senior Member Technical | ADP Pvt Ltd | Hyderabad, India | Nov 2019 – Mar 2022

Developed and maintained microservices components for ADP's enterprise payroll and HCM platform serving global clients.

- Built and owned microservices-based platform APIs using .NET Core and Azure, integrating multiple enterprise data sources and downstream systems.
- Designed event-driven workflows using Azure Service Bus and Azure Functions to automate payroll processing tasks, reducing manual operational overhead.
- Improved test coverage and CI/CD pipeline reliability, contributing to a more consistent delivery cadence across the team.

Tech Stack: .NET Core · Azure Functions · Azure Service Bus · Application Insights · SQL Server

Software Engineer | Option Matrix Infotech | Hyderabad, India | Oct 2016 – Nov 2019

- Built end-to-end ERP and CRM solutions using ASP.NET MVC, Angular, and SQL Server across multiple client engagements.
- Participated in Agile delivery cycles, authored unit tests, and contributed to code reviews.

KEY AI PROJECTS

1. Enterprise Advanced RAG Platform

A production-grade document intelligence platform built for legal, finance, and consulting use cases.

- Implemented hybrid retrieval (BM25 + dense vector search) with semantic re-ranking, significantly improving precision over pure vector search on long-form enterprise documents.
- Engineered chunking and embedding pipelines using sliding window and semantic chunking strategies tuned for financial and legal document structures.
- Built query rewriting and context enrichment stages to clarify and expand user queries before retrieval, improving answer groundedness and faithfulness.
- Extended the platform with an AWS Bedrock inference option, storing source documents in S3 and leveraging Bedrock Knowledge Bases for managed RAG — enabling side-by-side evaluation against the Azure AI Search retrieval path.
- Deployed via Azure Functions with API Management; integrated a RAGAS evaluation harness for ongoing retrieval quality measurement.

Tech Stack: Azure AI Foundry · Azure OpenAI · AWS Bedrock · AWS S3 · AWS Lambda · Azure AI Search · .NET 8 · Semantic Kernel · pgvector

2. Agentic AI Reporting System

An autonomous multi-agent orchestration system for enterprise financial reporting.

- Designed a three-agent pipeline: Data Retrieval Agent (SQL/API via MCP), Analysis Agent (LLM reasoning), and Report Generation Agent (structured document output).
- Built an MCP tool interaction layer enabling agents to query enterprise databases using natural language and invoke APIs dynamically at runtime.
- Implemented with Semantic Kernel + AutoGen; currently migrating to Microsoft Agent Framework (MAF) for enterprise lifecycle management.
- Agent state and conversation history persisted in AWS DynamoDB for low-latency, scalable session management across concurrent users.

Tech Stack: Semantic Kernel · AutoGen · MCP · Azure OpenAI · AWS Bedrock · AWS DynamoDB · .NET 8 · LangGraph

3. CareFlow AI-Powered Clinical Consultation System

End-to-end AI system automating clinical consultation workflows in a healthcare setting.

- Built an agentic clinical workflow engine handling patient intake, consultation documentation, and care plan generation using Azure OpenAI and Semantic Kernel.
- Implemented context-aware conversation management with persistent vector memory, maintaining clinical context across multi-turn patient-provider sessions.
- Developed a real-time alert and escalation layer via Azure Service Bus to route critical flags to appropriate care teams.

Tech Stack: .NET 8 · Azure OpenAI · Semantic Kernel · Azure Functions · Azure Service Bus · React

4. Financial Data Intelligence — XBRL/EDGAR Extraction & Anomaly Detection

A proof-of-concept financial intelligence platform targeting SEC EDGAR filings.

- Built an XBRL extraction pipeline to retrieve, parse, and normalise structured financial data from SEC EDGAR at scale (Azure Functions + Service Bus + PostgreSQL + Redis + Blob Storage).
- Designed anomaly detection APIs using numerical cross-period validation and statistical benchmarking against industry peer data.
- Produced a full HLD/LLD design covering ingestion architecture, orchestration, and evaluation strategy.

Tech Stack: Azure Functions · Service Bus · PostgreSQL · Redis · Blob Storage · Python · .NET 8

EDUCATION

Master of Science — Artificial Intelligence & Machine Learning | **Liverpool John Moores University, United Kingdom** | *Expected 2026*

Currently Pursuing

Postgraduate Diploma — Generative AI & Applied Machine Learning | **IIIT Bangalore** | *2025*
CGPA: 3.85 / 4.0

Bachelor of Technology — Electronics & Communication Engineering | **JNTU Kakinada, India** | *2016*

CERTIFICATIONS

- Microsoft Certified: Azure AI Engineer Associate (AI-102) In Progress, Expected 2026
- Executive PG in AI/ML — IIIT Bangalore, 2025